Rodenticide Replacement Program

A five year, $50M coordinated research program to tackle rodent fertility control.

Developed through the BiTS program, a part of Renaissance Philanthropy. Interested parties can reach Mal Graham (project lead) at mal@valeresearch.org

The problem

Globally, rodents cause major economic, public health, and conservation damages, including:

- **Substantial yield losses in agriculture and contamination of food stores:** In Asia, chronic pre-harvest rice losses of 10% are common in traditional farming systems, with outbreaks driving losses well above 50% in affected areas.[1] The direct costs alone of the 2021 Australian mouse plague are estimated at around A$ 100 million in damages.[2] In Africa, a systematic review covering 32 countries found crop losses of 46% at seedling stage and 15% at maturity.[3]
- **Hundreds of millions of cases of rodent-borne diseases annually:** Rodents are a significant reservoir for diseases including leptospirosis, plague, and Lassa fever. Leptospirosis is estimated to cause 1.03 million cases per year[4] with substantial costs to wellbeing and economic productivity primarily among marginalized communities.[5] Indirect impacts of rodents are harder to quantify but likely substantial – exposure to mouse allergens, for example, is common in urban regions and associated with asthma morbidity in sensitized children.[6,7]
- **Hundreds of species at threat:** Invasive rodents have been implicated in 75 extinctions, with five species alone collectively threatening 420 more.[8] Rodents are often considered the number one conservation risk on islands, where they prey on ground-nesting seabirds and other important wildlife.

These problems persist despite widespread use of rodenticides because of the twin forces of immigration and compensatory survival. When rodent populations are reduced or eliminated in one area, rodents from adjacent areas move in (immigration). Reductions in population sizes also increase per-capita resource availability, allowing more juveniles to survive (compensatory survival). The result is rapid rebound: in a recent field experiment, rodenticide and intensive trapping reduced multimammate mouse populations by 74–92%, but populations rebounded to pre-treatment levels within six months.[9] Furthermore, the increased migration of rodents in response to rodenticides accelerates the spread of the diseases they carry. In Vancouver, for example, lethal rat control has been associated with a nearly 10x increase in the odds that surviving rats carry leptospirosis.[16]

To prevent these rebounds, rodenticides need to be administered repeatedly at large scales: all connected areas that provide suitable rodent habitat at once. But all available rodenticides — from metal phosphides to second-generation anticoagulant rodenticides (SGARs) — are highly toxic, and it just isn't safe to administer rodenticides at these levels. Rodenticides cause:

- **Direct threats to human health**: Metal phosphides are a particularly common rodenticide in low- and middle-income countries, where access to SGARs is limited and extensive illegal rodenticide sales occur. Products like zinc and aluminum phosphide work by creating phosphine gas, which is deadly to humans, companion animals, and livestock and has no antidote. In India, for example, 54% of intentional or accidental ingestions are fatal,[17] likely causing thousands or tens of thousands of human deaths each year.

- **Harms to wildlife:** SGARs, the most common rodenticide type, linger in rodents and poison animals who eat them — undermining natural rodent control and exacerbating the very problem rodenticides are meant to address. Despite existing restrictions on SGAR use, clinically significant levels of exposure have been found across a wide range of non-target wildlife,[10-13] harming or killing valuable and rare species. The effects get worse the higher up the food chain you go: in the US, rodenticides have been found in the livers of 82% of bald eagles[14] and 95% of cougars.[15] These exposures are clinically significant, causing issues ranging from lethargy, nosebleeds, broken blood vessels, secondary diseases like mange, and internal hemorrhages.

The solution

The Rodenticide Replacement Program seeks to fund a portfolio of approaches to develop alternatives to rodenticides which are safe to apply at scale, attaining effective rodent population control without the harmful externalities of current approaches. **Fertility control is the best bet for an innovation that can meet the moment due to multiple promising lines of discovery**. First, the use of the lab rat (a domesticated brown rat, one of the most problematic rodent species) as a model organism means that almost every product explored for human contraceptive use has been tested in rats — and although many compounds face challenges translating from rodents to humans, for our program, rodents *are* the target. Second, developments in vaccination delivery in human medicine and self-spreading rabies vaccinations for wildlife may also be relevant to species-specific immunocontraceptives for rodents.

A complete solution includes the combination of an active ingredient, a pharmacokinetic pathway (getting a consumed active ingredient to the target organs), and a remote delivery system. The active ingredient should be highly efficacious, reducing fecundity by at least 80% for at least three months, to last the duration of a typical cropping season. Furthermore, the ideal solution must be effective in field conditions and safe for nontarget exposures (ideally, it should be species-specific).. Remote delivery must be regularly feasible over large regions to achieve the lasting impacts that have eluded rodenticides. While challenging, existing wildlife management programs offer useful precedents. Contraceptives have been successfully used to manage pigeons, horses, and elephants, and aerial administration of rabies vaccines has been extraordinarily successful.[18]

Our aim is to develop fertility control products that outperform rodenticides across many deployment contexts, from agriculture to island conservation. But even if we fall short of that target, fertility control will be able to complement rodenticides, affecting individual rodents that have learned to avoid rodenticides and offering a safe tool for use in areas with children, refugees, and other vulnerable populations. And the technologies we develop in this program could have broader impacts, translating to the vaccination of rodents to prevent or reduce zoonotic disease spread, the control populations of other invasive species, and the mitigation of human-wildlife conflicts that are caused by high animal population densities.

Why now?

The demand signal for rodenticides alternatives is already here. California and British Columbia have enacted bans on some classes of rodenticides, and other U.S. jurisdictions are contemplating ending exemptions that have allowed rodenticides to bypass rules applied to other pesticides. Most strikingly, the EU has designated SGARs as “candidates for substitution” – meaning that when a safe and effective alternative is available, these rodenticides will no longer be approved for use. This creates an enormous opportunity to bring innovation to the rodent control industry.

One company, SenesTech, has commercialized rodent fertility control products in the U.S. since 2016. However, their products are widely considered ineffective and insufficiently tested; the company's commercial trajectory suggests limited capacity for further R&D that could address the concerns. EP-1, a hormonal product, is in use to control some rodent species in China and Tanzania — but EU and US regulators have indicated they will not approve environmental administrations of hormone-based products, reducing commercial viability. The broader rodenticide industry, meanwhile, is not pursuing fertility control; conversations with industry contacts suggest they would be willing to adopt and produce an effective product, but they are averse to investing much in the required R&D themselves.

At the same time, the fertility control research community is systematically underfunded — too applied for NSF, too distant from human health for NIH, and too early-stage for commercial investment. The relevant innovations exist across widely separated research disciplines, from food science to human vaccine delivery. Without coordinated investment, the emerging bans are more likely to push rodent control backward toward historical options than forward toward better ones.

Our approach

We are seeking $50 million to advance a portfolio of approaches targeting landscape delivery of rodent contraception. Philanthropic investment will support parallel exploration of multiple delivery–cargo combinations, allowing us to concentrate funding toward the options that show the greatest potential.

Our program will focus on five species: the brown rat, the black rat, the ricefield rat, the house mouse, and the multimammate mouse. These species are responsible for the vast majority of agricultural, conservation, and human health issues worldwide. Candidate compounds will be tested in the lab to establish effectiveness, safety, and specificity. Successful compounds from this stage will then be tested in the field in controlled settings, with a target of advancing 2–3 compound-delivery pairings into field efficacy trials by month 36.

To combat the siloing that has historically slowed rodent fertility control development, we will begin with a six-month matchmaking process to bring teams together, forcing integration across reproductive biology, vaccine engineering, formulation chemistry, animal behavior, and field logistics. While the precise solution candidates will not be known until after this period, several technical developments make achieving our targets newly tractable.

Our team

Our team pairs fertility control domain expertise with deep operational experience in conservation and scientific program management. **Dr. Mal Graham** (program lead) and **Dr. Nitin Sekar** (co-lead) each bring a decade of experience designing and launching novel scientific programs and helping to build interdisciplinary programming from scratch. **Drs. Giovanna Massei, Jens Jacobs, Steven Belmain, and Gregg Howald** contribute to the program as technical advisors, bringing a combined 90+ years of rodent management and fertility control experience.

Appendix 1: Emerging and extant technologies enabling landscape-scale rodent fertility control

This program is newly tractable because of recent advances across three layers of the problem: the contraceptive cargo, the "micro" delivery (how the cargo gets to target organs e.g., through oral mucous membranes), and the "macro" delivery (how the product gets to the target animal; e.g., aerial broadcasting of an oral bait). Below we summarize several specific technical threads — none sufficient on its own, but each addressing a barrier that previously blocked progress.

Contraceptive cargo

- **Immunocontraceptive vaccines.** Vaccines that induce the immune system to attack components of the reproductive system can produce 1–2 years of infertility in target species. Existing immunocontraceptives are deployed in horses, deer, and prairie dogs, and laboratory work has demonstrated several promising candidates in rats. These products can be species-specific, nontoxic, sustainable, and long-lasting. The principal remaining barrier is route of administration: current products require injection, and successful conversion to oral delivery has not yet been achieved. Recent advances in oral vaccine formulation (below) make this barrier increasingly tractable.

- **Chemosterilants.** These are chemicals that, via a range of pathways, ultimately cause sterility. The existing compound used in rodent control is VCD, or 4-vinylcyclohexene diepoxide – but unfortunately this appears to be insufficiently effective, only inducing around 60% reduction in fertility in recent lab tests (unpublished). But at least one additional option, derived from compounds found to cause reduced fertility in humans, is being explored and shows a high degree of promise in the lab.

Micro-delivery mechanisms: once consumed by a rodent, the active ingredient must reach the bloodstream or other target organs. The principal barrier is that immunocontraceptive vaccines are typically degraded by the digestive tract before they can elicit an immune response. Two recent classes of solution are emerging:

- **Oral delivery that bypasses the gut**: Mucoadhesive formulations could allow gradual absorption via the mucous membranes, and microabrasive formulations could enable transient permeabilization of the buccal tissue for direct uptake into the bloodstream.

- **Encapsulations that shelter the active ingredients**: Nanoencapsulation systems can protect the vaccine through gastric transit. This protection reduces degradation by protecting the vaccine from gastric acid and other harsh aspects of the gastrointestinal tract, facilitates crossing of the intestinal epithelium, enhances the likelihood of immune activation, and reduces the likelihood of rapid hepatic clearance.

- **Pollen grains**: Natural microcapsules like pollen grains can be chemically processed and loaded with vaccine antigens or chemosterilants for intranasal or oral delivery. In mouse models, chemically processed pollen has successfully triggered innate immune responses through several pathways. The shells naturally adhere to intestinal tissues, prolonging interaction and enhancing delivery effectiveness. The main barriers to advancement appear to be identifying the appropriate cargo and moving the technology from the lab to the field.

Macro-delivery approaches:

- **Aerial broadcast of oral baits.** Oral baits are the workhorse delivery mechanism for rodenticides and large-scale wildlife vaccination campaigns. Of the latter, the largest to date has targeted rabies in European wildlife, primarily animals like foxes. These programs involved aerially distributing oral vaccines at huge scales. While the case is more complex for fertility control, as we would want to prevent non-target wildlife from losing fertility, the model may prove viable for any species-specific, orally deliverable compounds we are able to discover.

- **Transmissible or transferable vaccines.** Transmissible vaccines transfer between individuals, usually carried by a modified virus. Transferable vaccines are those which transfer from the treated animal to other animals, but no further: animal A transfers the vaccine to any animal B it encounters, but B does not transfer further to C. This is typically achieved through a paste or topical formulation that target animals groom from one another. While transmissible vaccines as an immunocontraceptive shouldn't be ruled out *a priori*, and have already been developed once for mice in Australia[20] (although never tested), there are substantial biosecurity issues that would need to be navigated.[21] Transferable vaccines likely represent a stronger option. As proof of concept, a paste-formulated rabies vaccine has recently been developed for this purpose in bats,[22] and tests with inactive markers show a promising degree of transfer.[23]

- **Tracking powders.** The transferable concept extends beyond vaccines. Tracking powders — applied to rodent runs and burrows, then ingested when animals groom themselves — are an established mechanism for population control, but current formulations are too acutely toxic for use in most contexts due to risks to humans, pets, and non-target wildlife. The delivery mechanism is well-validated; what is missing is a cargo with the safety and species-specificity profile this program targets. A grooming-activated fertility control compound would inherit a proven delivery pathway while minimizing the off-target hazards of current rodenticide powders.

No single one of these threads is sufficient on its own. The program's distinctive contribution is integration: pairing a viable contraceptive cargo with a delivery mechanism that reaches landscape scale, in a formulation suitable for field conditions. The most promising combinations — for example, a nanoencapsulated vaccine delivered via oral bait, or a grooming-activated immunocontraceptive paste — cut across research communities that currently operate independently. The six-month discovery phase will identify which compound–microdelivery-macrodelivery combinations to advance into laboratory and field testing.

Appendix 2: Team & Advisors

Team

Dr. Mal Graham (program lead) brings a decade of experience designing and launching novel scientific programs and helping to build interdisciplinary fields from scratch. As Executive Director of Wild Animal Initiative, they conceptualized and launched the organization's $3.5M scientific grantmaking program and grew the broader organization from 5 to 20 staff and $300K to $6M over five years; they also designed the inaugural grantmaking process for Arthropoda Foundation, where they currently serve as board treasurer. Their work in wild animal welfare science — itself a field built by integrating ecology, ethology, and animal welfare science — has involved launching new research areas through targeted workshops and matchmaking processes, the same model this program will deploy at larger scale. Mal received a PhD in Engineering Mechanics from Virginia Tech and a Bachelors of Physics and Philosophy from the University of Oxford.

Dr. Giovanna Massei (lead technical advisor; Botstiber Institute for Wildlife Fertility Control (BIWFC) & University of York) brings 30 years of wildlife management experience to the program, with 20 spent focusing entirely on wildlife fertility control research. She is the Europe Director for the BIWFC and a professor in the Department of Environment and Geography at the University of York, UK. Previously, she served as Senior Ecologist at the UK Animal and Plant Health Agency's National Wildlife Management Centre.

Dr. Nitin Sekar (co-lead; Director of Coexistence, Conservation X Labs) has 18 years experience in sustainable development, international conservation, and multidisciplinary research including monitoring and evaluation. He has spent eight years starting and developing new programs, as well as two years as a science-policy fellow at USAID. As WWF-India's first national lead for elephant conservation, Nitin knit together disparate regional programs and developed a nationwide strategy and research agenda, co-launching a manual for managing human-elephant conflict with the Government of India. He currently leads CXL's open innovation programs on rodenticide alternatives and alternative proteins. Nitin received a PhD and a certificate in science, technology, and environmental policy from Princeton University.

Advisors

Dr. Steve Belmain (NRI, University of Greenwich) is a leading international expert on rodent ecology, with research spanning the UK, Europe, Asia, and Africa focused on rodents as agricultural pests and disease vectors. He is a long-standing advocate for ecologically-based rodent management, developing sustainable control methods that integrate environmental, behavioural, and zoonotic risk factors. His work has been central to understanding both zoonotic transmission risks and the drivers of rodent population outbreaks, and contributed to the University of Greenwich's 2019 Queen's Anniversary Prize for pest management research.

Dr. Jens Jacob (Julius Kühn-Institut, Germany) leads the Rodent Research Group at the Julius Kühn-Institut – Federal Research Centre for Cultivated Plants. His work focuses on rodent ecology, population dynamics, and the development of ecologically and economically sustainable management methods. He has been directly engaged in advancing fertility control as a rodent management tool, co-authoring a 2024 framework for researchers and practitioners on developing fertility control for rodents and serving on the Botstiber Institute for Wildlife Fertility Control advisory committee. He also serves as vertebrates subject editor for *Pest Management Science* and the *Journal of Plant Diseases and Protection*, and chairs the German Phytomedical Society's vertebrate working group.

Gregg Howald (Coastal Conservation) has worked on invasive species eradication across 75+ islands in eight countries, formerly as Director of Global and External Affairs at Island Conservation and now through Coastal Conservation, which he cofounded — ensuring our research outputs are designed against the operational realities of island deployment.

Richard Parr MBE (Center for Wild Animal Welfare) worked as Special Adviser to the UK Prime Minister from 2012-2016, and as Special Adviser to the Secretary of State for International Development from 2010-2012 and 2016-2018. In government, his main focus was on international development policy, and he worked closely on the formation of the UN Sustainable Development Goals. From 2019-2023, he was Managing Director of the Good Food Institute Europe, a non-profit that works on alternative proteins like plant-based and cultivated meat. In his current role, Mr. Parr focuses on policy development to support animal welfare in the UK. Mr. Parr will contribute policy expertise to the team to ensure that developed compounds have a strong chance of gaining regulatory approval.

Dr. Amanda Matthes (Amodo Design) is the Animal Welfare Technology Lead at Amodo Design. She will connect pharmacological teams with delivery engineers, addressing the formulation–delivery integration that is among the program's central technical risks. Dr. Matthes holds a PhD in computer science from the University of Oxford with the CDT in Autonomous Intelligent Machines and Systems (AIMS CDT) and an undergraduate degree in physics from the University of Heidelberg.

Dr. Kevin Esvelt (MIT Media Lab) leads the Sculpting Evolution group within the MIT Media Lab, and has been a vocal advocate for biosecurity. Dr. Esvelt will provide biosecurity review, particularly for any candidate compounds involving transmissible or self-disseminating vectors — an area where his work on responsible development frameworks for self-spreading biotechnologies is directly applicable.

The program leadership additionally draws on an informal network of experienced research program managers, who advise on program design and stage-gate decisions. We are formalizing additional advisory relationships with researchers in vaccine delivery, formulation chemistry, and reproductive pharmacology during the program's discovery phase.

Appendix 3: Program structure

The below table outlines our draft program structure. This structure will be refined in consultation with our technical advisors.

Summary of activities & programs

Phase	Compound & Delivery teams	Ecology, Modeling & Validation teams	Budget
0	Discovery & matchmaking: convene relevant experts across a range of disciplines, such as vaccinologists, bait engineers, viral vector specialists, immunologists, rodent behavior ecologists, commercialization specialists, and end users (ag, urban, island conservation). Build program awareness, run workshops to build the teams, finalize evaluation framework around stability, safety/specificity, and scale. Issue paired-team RFP. **Entry criteria:** credible paired team with plausible technical thesis addressing at least one of the three S's as its distinctive contribution.	Ecology and modeling: using existing data and field work to fill gaps, develop a complete ecological picture of the ecology of reproductive rebounds and needed efficacy to develop a target product profile (needed reach, fertility effect, etc.) for each of the five target species across relevant contexts (agriculture, public health, and conservation). Validation team establishes shared infrastructure: standardized efficacy assays, regulatory navigation, testing facility partnerships, IP/data framework.	$5M
1	4–6 teams advance their fertility control approach in the lab, focusing on fertility, behavior, and welfare effects as well as any potential environmental concerns. Teams will target high efficacy at the individual level, with specific thresholds set by the team in collaboration with the modeling team that are calibrated to their deployment context and scale. Teams will simultaneously develop delivery approaches matched to their compound's stability and specificity constraints.	Modeling subteam builds population dynamics models for each team's target context using ecological data from the field. Models quantify the efficacy × reach × duration × area combinations required for population-level impact — informed by known rodent dynamics from phase 0.	$15M
Gate 1: Advance teams who can show a credible path to realistic target end goals for stability, specificity, and scale specific to their approach and context, affirmed by the modeling and validation teams.			
2	Teams refine formulations based on delivery constraints. Selected delivery teams scale prototypes; species-specificity validated in controlled non-target exposure studies.	Modeling team refines predictions using Phase 1 empirical data and designs mesocosm assessment protocols calibrated to each team's defined targets. Finalizes measurement approach for the three S's under field-relevant conditions.	$20M
Gate 2: Teams must demonstrate progress at ≥50% of their Phase 1-defined targets across stability, safety/specificity, and scale. Teams clearly below threshold cut; resources reallocate to leading candidates.			
3	Final 1–2 ingredient-delivery combinations administered in mesocosm trials. Winning team(s) may receive FRO-scale follow-on funding if work warrants concentration beyond the fund's horizon. Delivery platforms tested at ecologically relevant scale against the three S's criteria established with modeling team.	Mesocosm experiments evaluate candidates against Phase 2 success criteria. Results inform regulatory submissions and go-to-market strategy.	$10M
Gate 3: Leading candidates must demonstrate ≥80% of their defined targets for stability, safety/specificity, and scale under mesocosm conditions. Meeting this threshold triggers commercial handoff.			

Appendix 4: Broader impacts

We believe the rodent fertility control program will do more than just help with rodent control. First, as wildlife habitat is steadily replaced by human-modified landscapes, humans and animals are increasingly brought into conflict over space, food, and other resources. Simultaneously, societal attitudes increasingly oppose lethal management, lethal management alone is not always effective, and lethal management cannot be used in every context. We need a robust and energetic fertility control research community to be prepared to meet these challenges.

Second, advances in vaccine delivery that we expect from this program could translate into the wildlife health sector more broadly, with implications for human health. For diseases where a human vaccine is not possible, or is too expensive, vaccinating animals could offer the most tractable path to reducing human disease burden — rabies oral vaccination of wildlife reservoirs is the proof of concept, and similar approaches are being explored for Lyme disease, leptospirosis, and bovine tuberculosis. The delivery innovations developed here — particularly in bait stability, species-specific uptake, and transmissible or self-disseminating vectors — could be directly portable to these applications.

Third, the program will build durable scientific infrastructure. The matchmaking process and subsequent multi-year collaborations will create lasting links between research communities (reproductive biology, vaccine engineering, wildlife ecology, delivery chemistry) that currently operate in isolation. Even compounds that fail our specific criteria may yield tools, methods, or trained researchers that advance adjacent problems in wildlife management, invasive species control, and One Health.

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